

Python 1 Week Assignment Plan (Easy - Medium - Hard)

DAY 1 - Basics, Input Output, Conditions

Easy:

1. User Profile Generator

- Take name, age, city, profession input
- Display formatted profile

2. Simple Calculator

- Add
- Subtract
- Multiply
- Divide

Medium:

Student Grade System

- Take subject marks
- Calculate total
- Calculate percentage
- Assign grade

Hard:

ATM Machine Simulation

Features:

- Check balance
- Deposit money
- Withdraw money
- Exit

Rules:

- No negative amount
- Cannot withdraw more than balance

DAY 2 - Loops and Strings

Easy:

Number Analyzer

Check:

- Even/Odd
- Positive/Negative/Zero

Multiplication Table Generator

Medium:

Password Validator

Rules:

- Minimum 8 characters
- Uppercase
- Lowercase
- Number
- Special character

Hard:

Bank Login System

- Username password validation
- Only 3 attempts
- Account lock

DAY 3 - Lists, Tuple, Dictionary

Easy:

Shopping Cart

Features:

- Add item
- Remove item
- Show cart

Medium:

Employee Management System

- Add employee
- View employee
- Search by ID

Hard:

Student Database

- Store students

- Calculate total
- Average marks
- Highest scorer

DAY 4 - Functions and Modules

Easy:

Calculator using functions

Medium:

Currency Converter

Hard:

Expense Tracker

Features:

- Add expense
- View expenses
- Total spending
- Highest expense

DAY 5 - File Handling and Exceptions

Easy:

Notes Application

- Write notes
- Read notes
- Delete notes

Medium:

Login System using File

- Register
- Login

Hard:

Bank Transaction Logger

Save all transactions in file

DAY 6 - Object Oriented Programming

Easy:

Student Class

Properties:

- Name
- Roll number
- Marks

Methods:

- Display
- Calculate grade

Medium:

Employee Salary System

Concept:

- Inheritance
- Method overriding

Hard:

Library Management System

Classes:

- Book
- User
- Library

DAY 7 - Projects

Easy Project:

Contact Management System

Features:

- Add contact
- Update contact
- Delete contact
- Search contact

Medium Project:

Quiz Application

Features:

- Questions
- Score
- Result

Hard Project:

Mini E-Commerce Backend

Classes:

- Product
- Customer
- Order
- Payment

Features:

Admin:

- Add Product
- Update Product
- Delete Product

Customer:

- View Product
- Cart
- Checkout
- Generate Bill

Skills Covered:

Python Basics

Logic Building

Data Structures

Functions

File Handling

Exception Handling

OOP

Projects